Semiconductor Fundamentals

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Semiconductor Fundamentals (A11962)

Short Title:	Semiconductor Fundamentals		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Advanced

Description of Module / Aims

The course aims to introduce the student to the fundamentals of solid state theory in support of the analysis and design of semiconductor materials and devices. A comprehensive review of the design, operation and fabrication of schottky, junction and optoelectronic diodes is undertaken. Semiconductor device design and process simulation tools are also concurrently introduced.

Programmes

`SEMF-0001`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	3 / 5 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	3 / 5 / M
`SEMF-0001`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	3 / 5 / M

Indicative Content

- Fundamental Semiconductor Theory
- The Schottky Diode
- The Junction Diode
- Optoelectronic Devices
- Semiconductor TCAD Tools

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Illustrate the fundamental concepts of solid state theory as applied in the analysis of carrier transport in semiconductor materials.
2. Explain the fundamental physical processes employed in the fabrication of semiconductor diode structures.
3. Demonstrate sound knowledge of the fundamentals of design, operation and fabrication of schottky, junction and optoelectronic diode devices.
4. Apply the knowledge and comprehension gained in evaluating and relating macroscopic semiconductor diode performance parameters to semiconductor solid state theory and associated fabrication processes.
5. Utilise the tools and methods employed in carrier transport analysis and process simulation for semiconductor diode structures.

Learning and Teaching Methods

- Lectures
- Practicals / Project Work

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	
Practical	30%	5

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Muller, R. _Device Electronics for Integrated Circuits_. USA: Wiley, 2003.
- Pierret, R. _Semiconductor Device Fundamentals_. USA: Addison Wesley, 1996.
- Sze, S. _Physics of Semiconductor Devices_. USA: Wiley, 2006.

Supplementary Material

- Neamen, D. _Semiconductor Physics & Devices_. USA: Irwin, 1997.
- Tyagi, M. _Introduction to Semicondu__ctor Materials & Devices_. USA: Wiley, 1991.

Requested Resources

- ENGINEERING LAB: Electronics